

ORIGINAL ARTICLE

Atrial Fibrillation in Genotyped Dilated Cardiomyopathy: Epidemiology, Risk Factors, and Outcomes: Insights From the SHaRe Registry

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BACKGROUND: Atrial fibrillation (AF) is the most common arrhythmia in dilated cardiomyopathy (DCM) and is associated with adverse outcomes. However, the genetic determinants of AF risk and its prognostic significance across genotyped DCM subtypes remain unknown.

METHODS: In this observational cohort study, we analyzed 3117 genotyped patients with DCM from the SHaRe (Sarcomeric Human Cardiomyopathy Registry). AF prevalence, incidence, and clinical characteristics were assessed by genotype. Patients were classified as genotype-positive [G(+)] if they had a pathogenic or likely pathogenic variant in a DCM-associated gene, and genotype-negative [G(−)] if genetic testing was negative. Factors associated with incident AF were evaluated using Cox regression. AF was modeled as a time-dependent variable to evaluate associations with clinical outcomes.

RESULTS: Among 3117 genotyped patients with DCM (mean age, 48±15 years; 39% were female; and 35% were G[+]), 12.3% (n=384) had prevalent AF. Of 2491 patients without prevalent AF and with follow-up, 312 (12.5%) developed AF during a median of 4.5 years (interquartile range, 1.6–9.0). Among 2851 patients with follow-up, cumulative AF prevalence was 23.6%. *LMNA* had the highest AF incidence (7.6/100 patient-years) and cumulative AF prevalence (56.7%) and was the only genotype independently associated with incident AF compared with that in G(−) patients (hazard ratio, 5.52 [95% CI, 3.84–7.95]; *P*<0.001). *TTN* had an AF incidence of 2.1 per 100 patient-years and cumulative AF prevalence of 24.4%, comparable to that in G(−) patients. Older age, male sex, and prior heart failure hospitalization were also independently associated with incident AF. AF was independently associated with a higher risk of the composite clinical outcome (hazard ratio, 1.58 [95% CI, 1.29–1.94]; *P*<0.001), including heart failure, ventricular arrhythmias, and all-cause mortality.

CONCLUSIONS: In this large genotyped DCM cohort, AF burden and incident AF risk varied across genotypes. Only *LMNA* was associated with an increased risk of incident AF. AF onset was independently associated with adverse outcomes, supporting genotype-guided AF surveillance and management in DCM.

Key Words: atrial fibrillation ■ cardiomyopathy, dilated ■ genetics ■ heart failure ■ hospitalization

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WHAT IS NEW?

- Atrial fibrillation (AF) risk varies by disease-associated gene in dilated cardiomyopathy, with *LMNA* conferring the highest burden and greatest risk of incident AF.
- In *LMNA* heart disease, AF often precedes left ventricular systolic dysfunction.
- AF is independently associated with heart failure, malignant ventricular arrhythmias, and all-cause mortality beyond genetic and clinical risk factors.

WHAT ARE THE CLINICAL IMPLICATIONS?

- Genetic test results provide additional clinical context for tailoring AF surveillance in dilated cardiomyopathy.
- In *LMNA* cardiomyopathy, the high AF burden supports more intensive rhythm surveillance and earlier assessment of disease progression after AF onset.
- Across genotypes, AF identifies patients at higher risk of adverse events, supporting closer clinical surveillance.

Nonstandard Abbreviations and Acronyms

AF	atrial fibrillation
BMI	body mass index
DCM	dilated cardiomyopathy
G(+)/G(–)	genotype-positive/genotype-negative
HF	heart failure
LA	left atrial
LVEF	left ventricular ejection fraction
P/LP	pathogenic/likely pathogenic
SHaRe	Sarcomeric Human Cardiomyopathy Registry
VA	ventricular arrhythmia

Atrial fibrillation (AF) is the most common arrhythmia in dilated cardiomyopathy (DCM) and is associated with increased morbidity and mortality, including heart failure (HF) and stroke.^{1–5} Traditional clinical risk factors such as older age, higher body mass index (BMI), diabetes, and hypertension are established independent contributors to AF risk in the general population and in patients with DCM.^{2,6} However, studies specifically examining the factors associated with AF within genetically characterized DCM are limited. Prior studies have also largely treated DCM as a homogeneous disease entity, limiting insight into genotype-specific differences in AF risk.^{1,2,7,8}

With the growing integration of genetic testing into clinical cardiovascular care, the genetic underpinnings of arrhythmogenicity in DCM are increasingly recognized, providing much-needed granularity to the broad

classification of “nonischemic DCM.” Genotype-specific natural history studies have shown that genetic subtypes of DCM differ in clinical course and arrhythmia burden,^{9–12} suggesting that the consequences of AF may also vary by genotype. For example, *LMNA* variants are associated with early-onset AF with rapid progression from paroxysmal to persistent AF.¹¹ *TTN* truncating variants have been associated with more persistent AF subtypes and a greater need for repeated cardioversions compared with genotype-negative patients.¹³ In contrast, AF occurs only in a minority of patients with *FLNC* variants and is not considered a phenotype-defining feature for this disease gene.¹²

Although these studies provide valuable genotype-specific insights, they are often underpowered and do not provide a comprehensive overview of the AF risk and associated outcomes across the spectrum of genetic DCM. As such, there remains an unmet need to understand the burden, associated factors, and prognostic implications of AF in a clinically and genetically characterized DCM cohort.

To address this gap, we analyzed a genotyped DCM cohort from the international, multicenter SHaRe (Sarcomeric Human Cardiomyopathy Registry) to: (1) describe the incidence of new-onset AF and cumulative AF prevalence in DCM; (2) identify genetic and clinical factors associated with incident AF; and (3) evaluate the impact of AF on clinical outcomes.

METHODS

Patients were identified from the SHaRe-DCM consortium, an international, multicenter, longitudinal registry from high-volume DCM referral centers with detailed phenotypic, genetic, and clinical outcome data.

This study analyzed data collected until January 2025 from 11 international sites: Brigham and Women’s Hospital, MA, United States; Children’s Hospital of Philadelphia, PA, United States; Northwestern University, IL, United States; Stanford University Medical Center, CA, United States; University of Colorado Anschutz Medical Campus, CO, United States; University of Michigan, MI, United States; University of Pennsylvania, PA, United States; Maastricht University Medical Center, Maastricht, the Netherlands; Royal Brompton Hospital, London, United Kingdom; Careggi University Hospital, Florence, Italy; and University of Trieste, Trieste, Italy.

Data harmonization was performed by Boston Advanced Analytics using Aurora software, with data updated quarterly from participating sites. Institutional review and ethics approval were obtained in accordance with applicable site policies. Primary data are unavailable due to site-specific ethical restrictions. Analytic methods are available on request.

Study Population

In SHaRe-DCM, DCM was defined as a clinically adjudicated phenotype of cardiomyopathy after exclusion of ischemic, syndromic, and inflammatory causes, irrespective of left ventricular ejection fraction (LVEF) or chamber dilatation. Patients with a

phenotype-positive diagnosis of DCM who underwent genetic testing were included. The following patients were excluded: (1) family history of DCM without a DCM phenotype; (2) personal or family history of arrhythmogenic right ventricular cardiomyopathy based on the 2010 Task Force Criteria¹⁴; (3) missing data on age at the first SHaRe site visit or at last follow-up; and (4) unclassified genetic diagnosis, defined as patients in whom variants were identified but had not been systematically assessed or classified.

The follow-up period was defined as the time from the first SHaRe site evaluation to either the last clinical encounter or first censoring event (ventricular assist device implantation, heart transplant, or death).

Genetic Testing and Classification

Genetic testing was performed at participating SHaRe sites through various clinical laboratories. Variants were classified according to each laboratory's American College of Medical Genetics and Genomics/Association for Molecular Pathology–based criteria at the time of testing.¹⁵

Nineteen genes with an established gene-disease association for DCM, based on Clinical Genome Resource (ClinGen) curation,¹⁶ were defined as DCM-associated genes. Genes with definitive or strong evidence for DCM included *ACTN2*, *BAG3*, *DES*, *DSP*, *FLNC*, *LMNA*, *MYH7*, *PLN*, *RBM20*, *SCN5A*, *TNNC1*, *TNNT2*, and *TTN*. Genes with moderate evidence included *ACTC1*, *JPH2*, *NEXN*, *TNNI3*, *TPM1*, and *VCL*.

Variants in these DCM genes were classified as pathogenic/likely pathogenic (P/LP), variant of uncertain significance, or negative, defined as benign/likely benign or no reportable variants. Patients were stratified into 3 genotype groups: (1) genotype-positive [G(+)] if they had ≥ 1 P/LP variant; (2) variant of uncertain significance if they had ≥ 1 variant of uncertain significance and no P/LP variants; and (3) genotype-negative [G(–)] if they only had benign/likely benign variants or no reportable variants. Hereafter, gene names (eg, *LMNA* and *TTN*) will refer to phenotype-positive patients with P/LP variants unless stated otherwise.

Definitions and Outcomes

Prevalent AF was defined as AF diagnosed at or before the first SHaRe site visit. Incident AF was defined as new-onset AF during follow-up among patients without prevalent AF and with at least 1 follow-up visit. AF incidence was reported as the event rate per 100 patient-years. "No AF" was defined as the absence of an AF diagnosis at baseline and throughout the follow-up period. Cumulative AF prevalence was defined as the proportion of patients with AF by the end of follow-up (ie, those with prevalent AF and at least 1 follow-up visit and those who developed incident AF). AF duration was defined as the time from AF diagnosis to the last follow-up visit or censoring, among patients with complete age data.

The following clinical outcomes were evaluated: (1) composite HF, defined as hospitalization for HF, outpatient treatment with intravenous inotropes, implantation of durable ventricular assist device, heart transplant, or HF-related death; (2) composite of ventricular arrhythmia (VA), defined as sustained ventricular tachycardia, ventricular fibrillation, appropriate implantable cardioverter defibrillator therapy, cardiac arrest, or sudden cardiac death; (3) all-cause mortality; and (4) overall

composite outcome, defined as the occurrence of any of the above outcomes.

Statistical Analysis

Continuous variables were summarized as mean (SD) or median (interquartile range) and compared using Student *t* test or Wilcoxon rank-sum test, as appropriate. Categorical variables were reported as counts and percentages and analyzed using χ^2 or Fisher exact test. Pearson coefficients were used to assess pairwise correlations.

Variants in non-*TTN* sarcomeric genes (*ACTC1*, *MYH7*, *TNNC1*, *TNNI3*, *TNNT2*, and *TPM1*) were aggregated into a single category for analysis due to their similar functional role in cardiac contractility. Genes with <30 individuals carrying P/LP variants (ie, *PLN*, *BAG3*, *RBM20*, *SCN5A*, *NEXN*, and *VCL*) were not analyzed individually in statistical models due to limited power. However, these genes were retained within the broader genotype-positive group and included in all grouped analyses.

For comparisons across multiple genotype groups, 1-way ANOVA was used for normally distributed variables, and the Kruskal-Wallis test was used for non-normally distributed variables. Prevalent AF was analyzed using multivariable logistic regression, adjusting for conventional clinical factors. AF incidence rates were compared across genotype groups using Poisson regression, with different reference groups depending on the comparison of interest (eg, G(–), *LMNA*, and *DSP*). Cumulative AF prevalence was compared using χ^2 tests for pairwise comparisons between genotype groups.

Baseline LVEF was based on transthoracic echocardiography performed within 1 year of the first SHaRe site visit. If transthoracic echocardiography was unavailable, cardiac magnetic resonance–derived LVEF was used when available. For Cox regression analyses, variables including weight, BMI, LVEF, left atrial (LA) diameter, indexed LA volume, and LV end-diastolic diameter exhibited missingness (Table S1). These variables were imputed using multiple imputation by chained equations with linear regression conditional models, in which all imputed variables were used to inform one another, along with other fully observed covariates, including age at first site visit, sex, genotype, and enrolling SHaRe site. Twenty imputed data sets were generated and combined using Rubin's rules.

Cox regression was used to evaluate associations between baseline variables and incident AF and, separately, the association between incident AF and clinical outcomes, with AF modeled as a time-dependent variable in the latter analysis. Both multivariable analyses used the same covariates. As a sensitivity analysis, competing risk regression using the Fine-Gray hazards model was performed for the composite HF and composite VA outcomes, treating non-HF death and nonsudden cardiac death, respectively, as competing events. All models were stratified by SHaRe site to account for center-level variation. The proportional hazards assumption was assessed for all Cox regression models. Statistical significance was set at a 2-tailed $P \leq 0.05$. All analyses were performed using Stata/BE 18.0 for Mac.

RESULTS

Of the 8503 individuals in the SHaRe-DCM consortium, 3117 patients met study inclusion criteria, representing the genotyped DCM cohort (Figure S1). At the time of

first SHaRe evaluation, 384 patients (12.3%) had prevalent AF. A total of 2851 patients had at least 1 follow-up visit and were eligible for longitudinal AF analyses, including 2491 patients without prevalent AF and 360 patients with prevalent AF. At the end of follow-up, 672 patients had AF, corresponding to a cumulative AF prevalence of 23.6% (Table S2).

During a median follow-up of 4.5 years (interquartile range, 1.6–9.0), 312 patients (12.5%) developed new-onset AF, corresponding to an incidence rate of 2 cases per 100 patient-years. A total of 2179 patients remained free of AF throughout the study period.

Baseline Characteristics by AF Status

Characteristics of the study cohort at the initial SHaRe site evaluation stratified by AF status are summarized in Table 1. Patients with AF were significantly older at DCM diagnosis and at first site visit than those without AF: 86% (n=331) of the prevalent AF group and 78% (n=243) of the incident AF group were aged ≥ 40 years, compared with 66% (n=1442) in the no AF group (both $P<0.001$). Both AF groups had a significantly lower proportion of females compared with the no AF group (25% and 31%, respectively, versus 43%, both $P<0.001$).

Both AF groups had a more adverse baseline clinical profile than patients without AF. Those with prevalent AF had higher rates of DM, hypertension, prior stroke, and history of HF hospitalization. In contrast, patients who developed incident AF consistently showed intermediate comorbidity profiles, with values between the prevalent AF and no AF groups. The average BMI across all 3 groups was 27.3 kg/m². Patients with prevalent AF had the highest proportion of obesity (BMI ≥ 30 kg/m²) at 33% (n=76), compared with 26% (n=373) in the no AF group ($P=0.021$). Obesity prevalence in the incident AF group (32%, n=50) was not statistically different from the no AF group ($P=0.089$). Overall, males had higher BMI (27.5 $\pm$ 5.3 versus 26.8 $\pm$ 6.9 kg/m²; $P=0.013$) and body weight (87.6 $\pm$ 19.3 versus 72.1 $\pm$ 20.1 kg; $P<0.001$) compared with females.

Among echocardiographic parameters, LA diameter showed the most pronounced association with AF status, being larger in the prevalent AF group (44 $\pm$ 9 mm) compared with those without AF (39 $\pm$ 8 mm; $P<0.001$). LA diameter was also significantly larger in males than in females (41 $\pm$ 8 versus 37 $\pm$ 7 mm; $P<0.001$).

G(+) status was more frequent in patients with AF: 39% in the prevalent AF group and 44% in the incident AF group, compared with 35% in the no AF group ($P=0.015$ and $P=0.002$, respectively). In a multivariable model adjusted for conventional risk factors, including age at first site visit, sex, diabetes, hypertension, prior HF hospitalization, and stroke, *LMNA* showed the strongest association with prevalent AF (odds ratio, 5.33 [95% CI, 3.69–7.69]; $P<0.001$), whereas *TTN* had a weaker

association (odds ratio, 1.47 [95% CI, 1.04–2.05]; $P=0.031$), and other genotypes were not significant.

Genotype-Specific AF Burden and Clinical Characteristics

Our analyses focused on the G(–) cohort and DCM-associated genes with at least 30 patients with a P/LP variant: *TTN*, *LMNA*, *DSP*, *FLNC*, and sarcomeric genes.

Both AF incidence and cumulative AF prevalence were higher in the G(+) group compared with the G(–) group. There was substantial heterogeneity in AF burden across genetic subgroups (Figure 1), with *LMNA* showing the highest AF incidence rate (7.6 per 100 patient-years) and cumulative AF prevalence (56.7%). *LMNA* was the only genotype with a significantly higher AF incidence rate than the G(–) group. Cardiac device use was also highest in *LMNA* (64%; Table S3). When *LMNA* was used as the reference group, all other genotypes, including G(–) status, had significantly lower AF incidence rates (all $P<0.001$). The *TTN* group had an AF incidence rate (2.1 per 100 patient-years) and cumulative AF prevalence (24.4%) similar to those of the G(–) group (1.7 per 100 patient-years and 20.6%, respectively), with no significant differences (incidence, $P=0.231$; prevalence, $P=0.107$). Notably, the *DSP* group, despite being the third-largest genotype subgroup (n=156), had the lowest AF incidence (0.8 per 100 patient-years) and prevalence (7.4%). This burden was significantly lower than that observed in the *LMNA*, *TTN*, and *FLNC* groups (all $P<0.05$ for both incidence and prevalence), and numerically lower than that observed in the sarcomeric variant group (incidence, $P=0.404$; prevalence, $P=0.313$).

Patients in the *LMNA* group were younger at AF diagnosis (mean, 47.6 $\pm$ 13.5 years) compared with those with *TTN* (52.6 $\pm$ 15.3 years, $P=0.034$) and G(–) (57.1 $\pm$ 13.4 years, $P<0.001$). In addition, among all the genotypes, only the *LMNA* group had a mean age at AF diagnosis (47.6 $\pm$ 13.5 years) that preceded the onset of DCM, defined by LVEF $<50\%$ (49.9 $\pm$ 12.7 years; Figure 2). In a pairwise analysis restricted to the *LMNA* group with complete age data for both AF and DCM (n=83), 75.9% of patients developed AF before the onset of systolic dysfunction, with AF occurring an average of 9.2 years earlier ($P<0.001$).

Echocardiographic parameters also showed genotype-specific differences among all groups (Table S4). The *TTN* group had the lowest mean LVEF (36 $\pm$ 14%, $P<0.001$), the largest LV end-diastolic diameter (59 $\pm$ 8 mm, $P<0.001$), and the largest LA diameter (41 $\pm$ 7 mm, $P<0.001$). In contrast, patients with *LMNA*, *DSP*, and *FLNC* had a relatively higher LVEF (all $\approx 47\%$) and smaller LV end-diastolic diameter dimensions ($P<0.001$). Patients with *LMNA* had a significantly higher baseline LVEF (47% $\pm$ 15%) than both *TTN* (36% $\pm$ 14%) and G(–) patients (37% $\pm$ 14%, both $P<0.001$). In addition,

Table 1. Characteristics of Dilated Cardiomyopathy Patients at Initial SHaRe Site Evaluation Stratified by Atrial Fibrillation Status

Variable	Prevalent AF (n=384)	Incident AF (n=312)	No AF (n=2179)	P value (prevalent AF vs no AF)	P value (incident AF vs no AF)
Demographics					
Age at DCM diagnosis, y	53 (14; n=381)	51 (14; n=302)	45 (16; n=2143)	<0.001	<0.001
Age at first site visit, y	55 (13)	51 (14)	46 (16)	<0.001	<0.001
Female, n (%)	96 (25)	96 (31)	934 (43)	<0.001	<0.001
Race*, n (%)	(n=232)	(n=146)	(n=1007)	0.519	0.939
White	212 (91)	130 (90)	889 (89)		
Black	15 (7)	11 (7)	77 (8)		
Asian	5 (2)	4 (3)	33 (3)		
Vitals at baseline					
Systolic blood pressure, mm Hg	117 (17; n=207)	123 (19; n=123)	120 (19; n=1230)	0.019	0.134
Diastolic blood pressure, mm Hg	74 (11; n=207)	76 (12; n=123)	74 (12; n=1229)	0.795	0.046
Heart rate, bpm	77 (18; n=160)	75 (17; n=72)	77 (19; n=667)	0.881	0.388
BMI, kg/m ²	28.3 (5.8; n=232)	28.0 (5.9; n=157)	27.1 (6.0; n=1459)	0.005	0.081
Weight, kg	88.6 (20.4; n=236)	85.6 (20.8; n=158)	80.3 (21.1; n=1472)	<0.001	0.003
Genetic diagnosis					
G(+)	149 (39)	135 (44)	757 (35)	0.007†	0.009†
G(−)	154 (40)	126 (40)	1057 (48)		
VUS	81 (21)	51 (16)	365 (17)		
G(+) vs G(−) only	...	...	...	0.015‡	0.002‡
Comorbidities					
Diabetes	65 (17)	28 (9)	130 (6)	<0.001	0.041
Hypertension	122 (32)	67 (21)	350 (16)	<0.001	0.017
Stroke	59 (15)	16 (5)	50 (2)	<0.001	0.004
Heart failure status					
Prior HF hospitalization	62 (16)	40 (13)	238 (11)	0.003	0.319
NYHA III/IV	50 (13)	24 (8)	164 (8)	<0.001	0.917
Cardiac Function					
LVEF§, %	37 (14; n=340)	40 (16; n=263)	38 (15; n=1949)	0.044	0.172
LVEDD, mm	58 (8; n=268)	58 (10; n=235)	58 (10; n=1725)	0.640	0.616
LA diameter, mm	44 (9; n=135)	40 (8; n=109)	39 (8; n=885)	<0.001	0.168

Values are n, n (%), or mean (SD). AF indicates atrial fibrillation; BMI, body mass index; bpm, beats per minute; DCM, dilated cardiomyopathy; G(+), genotype-positive with pathogenic/likely pathogenic variant in DCM genes; G(−), genotype-negative (benign/likely benign variant in DCM genes or no variant identified); HF, heart failure; LA diameter, left atrial diameter (in parasternal long-axis view); LVEDD, left ventricular end-diastolic diameter; LVEF, left ventricular ejection fraction; NYHA, New York Heart Association functional classification; TTE, transthoracic echocardiography; and VUS, variant of uncertain significance in DCM genes.

*Race not reported for patients with unclassified or missing data.

†Chi-square test *P* value compares all genetic diagnoses.

‡Chi-square test *P* value compares G(+) vs G(−); excludes VUS.

§LVEF based on TTE data within 1 year of the first site visit, or on CMR data if TTE data were unavailable.

||Statistically significant results.

patients with *LMNA* had a smaller LA diameter compared with the *TTN* group (39±8 versus 41±7 mm, *P*=0.028).

Factors Associated With Incident AF

Among 2491 patients without prevalent AF, 312 (12.5%) developed incident AF over a median follow-up of 4.5 years (interquartile range, 1.6–9.0; Table 2). The estimated cumulative incidence of AF was 10% (95% CI, 8.7%–11.5%) at 5 years, 16.8% (95% CI,

14.8%–19.0%) at 10 years, and 23.9% (95% CI, 21.0%–27.2%) at 15 years.

In univariable analyses, both older age at DCM diagnosis and at the initial site visit were consistently associated with an increased risk of incident AF, with a similar magnitude of association (hazard ratios [HRs], ≈1.3 per 10 years, both *P*<0.001). Male sex was associated with an ≈2-fold higher risk of incident AF. Patients with G(+) status had a higher risk of incident AF compared with those with G(−) status (HR, 1.54 [95% CI, 1.19–2.01]; *P*=0.001). Each 10 kg increase in body weight was

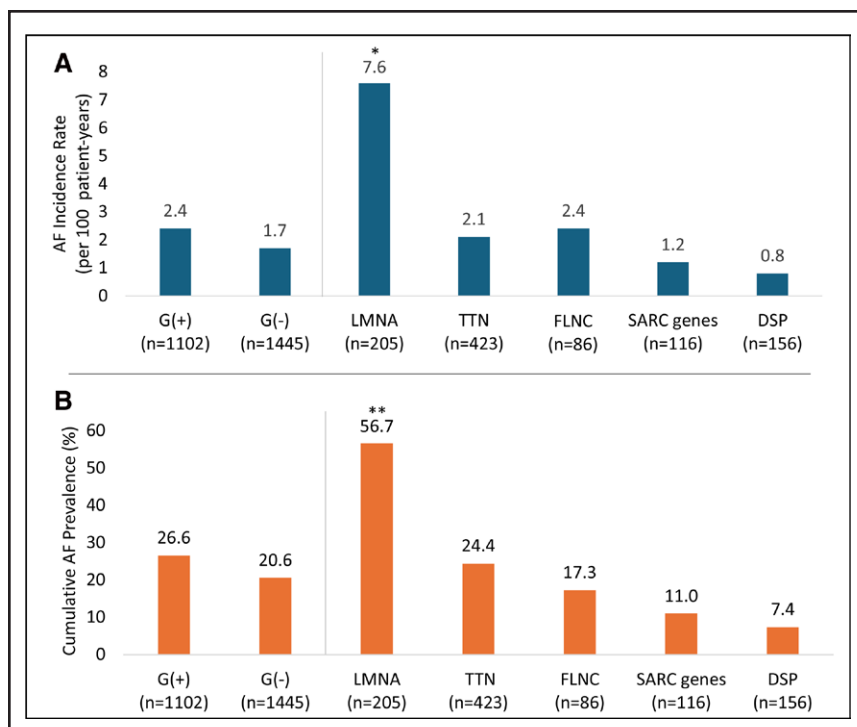


Figure 1. Atrial fibrillation (AF) burden in dilated cardiomyopathy (DCM) stratified by genotype.

(A) AF incidence rate per 100 patient-years. (B) Cumulative AF prevalence. Patients with *LMNA* had a significantly higher AF incidence (7.6 per 100 patient-years) and cumulative prevalence (56.7%) compared with patients with G(-) status, whereas no other genotypes showed significant differences compared with G(-) patients. G(+) indicates genotype-positive with pathogenic/likely pathogenic (P/LP) variant in DCM genes; G(-), genotype-negative (benign/likely benign variant in DCM genes or no variant identified); SARC genes, DCM-related sarcomeric genes with pathogenic/likely pathogenic variant (*ACTC1*, *MYH7*, *TNNC1*, *TNNI3*, *TNNT2*, and *TPM1*). * $P < 0.001$ for *LMNA* vs G(-), Poisson regression. No other genotype had a statistically higher AF incidence than G(-). ** $P < 0.001$ for *LMNA* vs G(-), χ^2 test. No other genotype had a statistically higher cumulative AF prevalence than G(-).

associated with a 10% higher risk of incident AF. Prior HF hospitalization (HR, 1.54 [95% CI, 1.09–2.15]; $P=0.013$), advanced New York Heart Association class (III/IV) (HR,

1.65 [95% CI, 1.06–2.56]; $P=0.027$), and LA diameter were significantly associated with incident AF risk (HR, 1.18 per 5 mm [95% CI, 1.09–1.29]; $P < 0.001$).

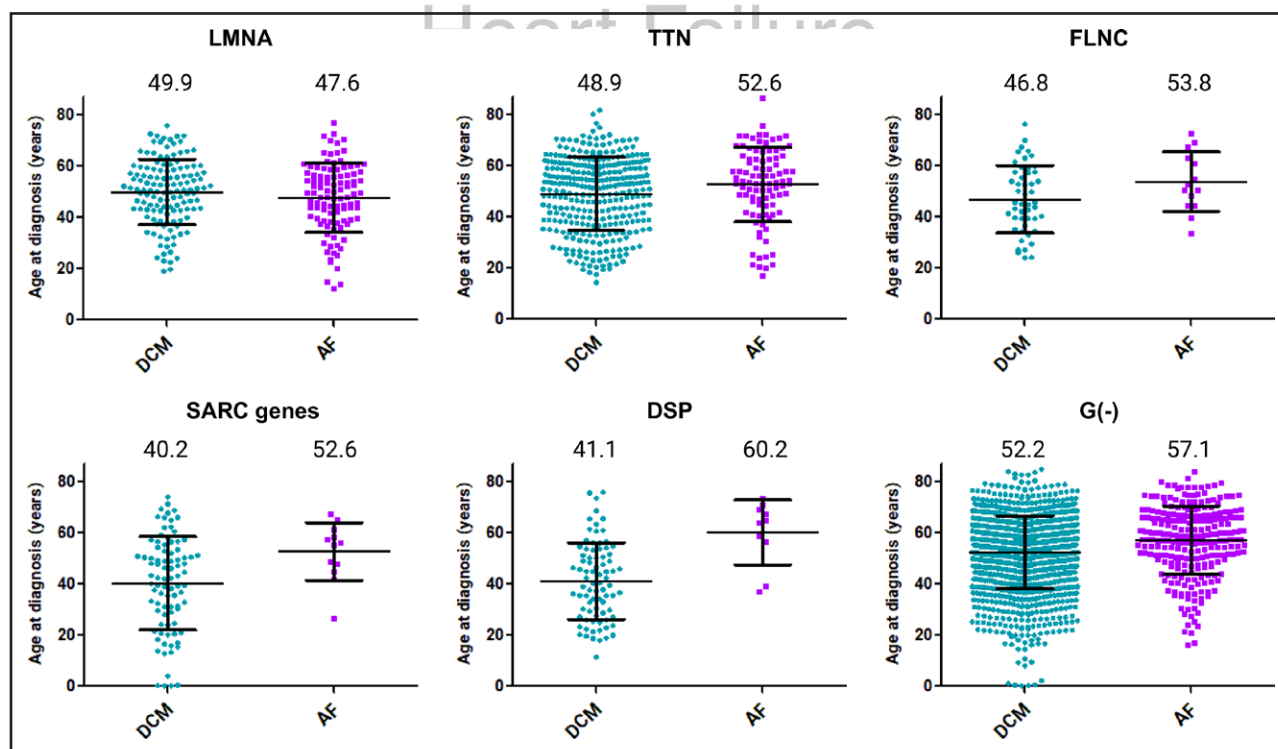


Figure 2. Comparison of age at diagnosis of atrial fibrillation (AF) and dilated cardiomyopathy (DCM) by genotype.

In *LMNA*, AF onset preceded left ventricular systolic dysfunction in the majority of cases (76%), whereas, in other genotypes, AF typically occurred after DCM diagnosis. Numbers above each plot represent the mean age at diagnosis (in years). Horizontal lines indicate the median and interquartile range. G(-) indicates genotype-negative (benign/likely benign variant in DCM genes or no variant identified); SARC genes, DCM-related sarcomeric genes with pathogenic/likely pathogenic variant (*ACTC1*, *MYH7*, *TNNC1*, *TNNI3*, *TNNT2*, and *TPM1*).

Table 2. Factors at Initial SHaRe Site Evaluation Associated With Incident Atrial Fibrillation in Dilated Cardiomyopathy

Variable	Univariable HR* (95% CI, P value)	Multivariable HR*† (95% CI, P value)
Age at diagnosis (per 10 y)	1.28 (1.18–1.39), $P<0.001$	...
Age at first site visit (per 10 y)	1.32 (1.22–1.44), $P<0.001$	1.35 (1.23–1.48), $P<0.001$
Male (ref: female sex)	1.95 (1.53–2.49), $P<0.001$	1.77 (1.35–2.33), $P<0.001$
Race		
Black (ref: White race)	1.03 (0.55–1.92), $P=0.926$	...
Asian (ref: White race)	0.77 (0.27–2.16), $P=0.615$	...
Systolic blood pressure (per 10 mm Hg)	1.03 (0.93–1.14), $P=0.566$	...
Diastolic blood pressure (per 10 mm Hg)	1.10 (0.94–1.27), $P=0.227$	...
Heart rate (per 10 bpm)	1.02 (0.87–1.18), $P=0.831$	...
BMI (per 5 kg/m ²)	1.07 (0.95–1.20), $P=0.267$	...
Weight (per 10 kg)	1.10 (1.04–1.18), $P=0.003$	1.05 (0.97–1.13), $P=0.243$
Genetic diagnosis (ref: genotype-negative)		
G(+)	1.54 (1.19–2.01), $P=0.001$	...
Genotype (ref: genotype-negative)		
P/LP <i>LMNA</i>	4.52 (3.19–6.42), $P<0.001$	5.52 (3.84–7.95), $P<0.001$
P/LP <i>TTN</i>	1.38 (0.97–1.96), $P=0.076$	...
P/LP <i>DSP</i>	0.52 (0.25–1.08), $P=0.079$	...
P/LP <i>FLNC</i>	1.52 (0.80–2.88), $P=0.204$	...
P/LP SARC genes‡	0.84 (0.44–1.62), $P=0.603$	...
G(+) [Other genes]§	1.22 (0.67–2.22), $P=0.520$	...
Diabetes (No vs yes)	1.40 (0.94–2.07), $P=0.095$	1.12 (0.75–1.68), $P=0.571$
Hypertension (No vs yes)	1.28 (0.97–1.69), $P=0.082$	1.07 (0.80–1.43), $P=0.636$
Previous stroke (No vs yes)	1.54 (0.93–2.57), $P=0.095$	...
Previous HFH (No vs yes)	1.54 (1.09–2.15), $P=0.013$	1.52 (1.05–2.20), $P=0.026$
NYHA III/IV (No vs yes)	1.65 (1.06–2.56), $P=0.027$	...
LVEF (per 5%)	0.98 (0.94–1.02), $P=0.374$	1.01 (0.96–1.06), $P=0.791$
LVEDD (per 5 mm)	1.02 (0.95–1.10), $P=0.512$	...
LA diameter (per 5 mm)	1.18 (1.08–1.29), $P<0.001$	1.05 (0.94–1.17), $P=0.376$

BMI indicates body mass index; DCM, dilated cardiomyopathy; G(+), genotype-positive with P/LP variant in DCM genes; G(–), genotype-negative (benign/likely benign variant in DCM genes or no variant identified); HFH, heart failure hospitalization; HR, hazard ratio; LA diameter, left atrial diameter; LVEDD, left ventricular end-diastolic diameter; LVEF, left ventricular ejection fraction; NYHA, New York Heart Association functional classification; P/LP, pathogenic/likely pathogenic; SARC genes, DCM-related sarcomeric genes with pathogenic/likely pathogenic variant; and VUS, variant of uncertain significance in DCM genes.

*Models were stratified by site to account for differences in baseline hazard.

†Adjusted for age at first site visit, sex, genetic diagnosis [G(–), G(+), *LMNA*, G(+) non-*LMNA*, VUS], weight, diabetes, hypertension, prior HF hospitalization, LVEF and LA diameter.

‡SARC genes includes *ACTC1*, *MYH7*, *TNNC1*, *TNNI3*, *TNNT2*, and *TPM1*.

§Other genes include P/LP variants grouped together due to small sample size (*BAG3*, *DES*, *RBM20*, *SCN5A*, *PLN*, *NEXN*, and *VCL*).

||Statistically significant results.

When this univariable model was expanded to include genotype-specific groups, only the presence of *LMNA* demonstrated a significantly increased risk of incident AF (HR, 4.52 [95% CI, 3.17–6.43]; $P<0.001$). Other genotypes, including *TTN*, *DSP*, *FLNC*, and sarcomeric genes, were not significantly associated with incident AF.

Prespecified clinically relevant covariates were evaluated using multivariable Cox regression models to assess associations with incident AF, with sequential models shown in Table S5. The covariates included age at first site visit, sex, genetic diagnosis [G(–), G(+), *LMNA*, G(+) non-*LMNA*, variant of uncertain significance),

weight, diabetes, hypertension, prior HF hospitalization, LVEF, and LA diameter. Both body weight and LA diameter were correlated with sex ($r=0.36$ and $r=0.27$, respectively), with men having greater body weight and a larger LA diameter. In multivariable analysis (Table 2), older age at first site visit, male sex, and prior HF hospitalization remained independently associated with increased risk of incident AF. *LMNA* had the strongest genetic association with an ≈ 6 -fold higher risk (HR, 5.52 [95% CI, 3.84–7.95]; $P<0.001$). In contrast, other genotypes were not associated with incident AF after adjustment.



Impact of AF on Clinical Outcomes

In univariable Cox regression models with AF modeled as a time-dependent covariate, AF was associated with significantly increased risk across all clinical outcomes. After adjustment for age, sex, genetic diagnosis, weight, diabetes, hypertension, prior HF hospitalization, LVEF, and LA diameter, AF remained independently associated with adverse outcomes. Specifically, AF was associated with a 58% higher risk of the overall composite outcome (HR, 1.58 [95% CI, 1.29–1.94]; $P<0.001$). This association was primarily driven by the HF-related composite end point (HR, 2.22 [95% CI, 1.71–2.88]; $P<0.001$) and all-cause mortality (HR, 1.82 [95% CI, 1.40–2.35]; $P<0.001$). AF was also associated with an increased risk of composite VA after adjustment, although the magnitude of effect was more modest and attenuated (HR, 1.36 [95% CI, 1.02–1.81]; $P=0.034$) relative to HF and mortality outcomes (Figure 3). Competing risk analysis yielded results consistent with the primary Cox models for both composite HF and VA outcomes (Table S6).

To explore genotype-specific differences in clinical outcomes after AF onset, we compared the cumulative incidence of the overall composite outcome among patients with AF who had *LMNA* or *TTN*, the 2 most prevalent genotypes in our cohort. In Cox models adjusted for age at first site visit, sex, diabetes, hypertension, prior HF hospitalization, and LVEF, patients with *LMNA* and AF had a higher risk of adverse events compared with those with *TTN* and AF (HR, 1.94 [95% CI, 1.08–3.49]; $P=0.027$).

DISCUSSION

In this large, multicenter, genotyped cohort of patients with DCM, we identified 4 key findings. First, within the

G(+) cohort, AF burden was substantial in patients with *LMNA*, with the highest incidence (7.6 cases per 100 patient-years) and a cumulative prevalence exceeding 50%. Second, *LMNA* genotype emerged as the strongest independent factor for developing incident AF, conferring >5-fold higher risk compared with G(–) patients, even after adjusting for clinical covariates. Third, despite more advanced structural remodeling, including larger LA size and lower LVEF, AF rates in the *TTN* group were lower than in the *LMNA* group and comparable to the G(–) group, underscoring the impact of genotype-specific differences in arrhythmogenic risk. Finally, AF was independently associated with adverse clinical outcomes across all end points after multivariable adjustment, with particularly strong associations for HF events and all-cause mortality, suggesting that AF is both a marker and a potential mediator of disease progression in DCM even after accounting for genotype. To our knowledge, this study represents the first comparative assessment of AF burden and prognostic impact across multiple genotyped DCM subtypes.

LMNA: Highest Burden and Incident Risk of AF in DCM

LMNA-related cardiomyopathy is characterized by high age-dependent penetrance, with electrical disease frequently preceding overt DCM.^{11,17} In our study, *LMNA* was the dominant genetic determinant of AF burden and risk, consistent with contemporary *LMNA* natural history studies reporting AF prevalence ≈60%.¹⁷ Furthermore, *LMNA* was also the only genotype in which AF onset preceded LV systolic dysfunction. Although the unpaired group means differed by only 2 years, paired analysis

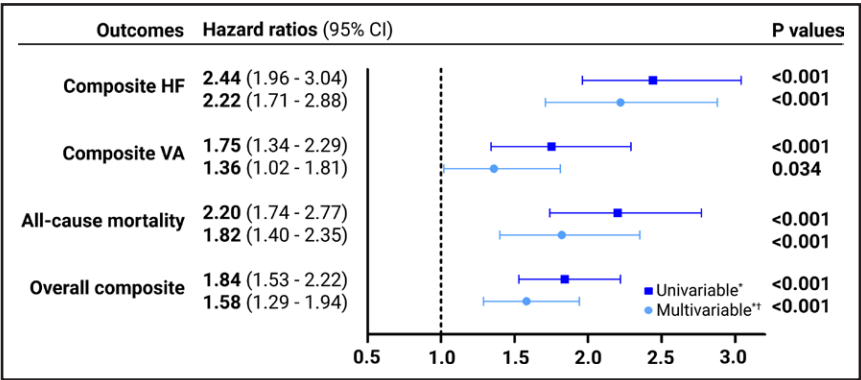


Figure 3. Impact of time-dependent atrial fibrillation (AF) on the incidence of clinical outcomes. This analysis illustrates that incident, time-dependent AF remains independently associated with adverse outcomes after accounting for genotypes and clinical covariates in genotyped dilated cardiomyopathy (DCM). Composite HF indicates composite heart failure (includes hospitalization for HF, outpatient treatment with intravenous inotropes, implantation of durable ventricular assist device, heart transplant, or HF-related death); Composite VA, composite ventricular arrhythmias (includes ventricular tachycardia, ventricular fibrillation, appropriate implantable cardioverter defibrillator [ICD] therapy, cardiac arrest, or sudden cardiac death); G(+), genotype-positive with pathogenic/likely pathogenic (P/LP) variant in DCM genes; G(–), genotype-negative (benign/likely benign variant in DCM genes or no variant identified); LA diameter, left atrial diameter; LVEF, left ventricular ejection fraction; and VUS, variant of uncertain significance in DCM genes. *Models were stratified by site to account for differences in baseline hazard. †Adjusted for age at first site visit, sex, genetic diagnosis [G(–), G(+) *LMNA*, G(+) non-*LMNA*, VUS], weight, diabetes, hypertension, prior HF hospitalization, LVEF, and LA diameter.

showed that AF onset occurred nearly a decade earlier in 76% of patients with *LMNA*, whereas, in other genotypes, AF occurred after DCM diagnosis. These findings suggest a primary atrial myopathy in *LMNA*, distinct from secondary atrial remodeling seen in other genetic and nongenetic DCM, and may help explain genotype-specific predisposition to AF.

Proteomic analysis of cardiomyocytes from patients with advanced *LMNA* undergoing heart transplant showed reduced *LMNA* protein expression in the atria but not in the ventricles,¹⁸ suggesting selective atrial vulnerability. Consistent with this, functional studies of a familial *LMNA* variant identified atrial nuclear abnormalities, impaired DNA repair, and contractile dysfunction.¹⁹ Electrophysiological mapping in patients with *LMNA* undergoing AF ablation further demonstrated extensive LA low-voltage areas and reduced interatrial conduction velocity, providing evidence of an abnormal atrial electrical substrate.²⁰ This atrial involvement is also evident in the subclinical stages, with reduced LA contractile strain observed before LA dilatation or LV dysfunction.²¹ Compared with patients with *TTN*, those with *LMNA* had more severe LA contractile dysfunction, even after adjustment for confounders.²¹ Together, these pathophysiological findings support an intrinsic atrial myopathy in *LMNA* and are consistent with our observation that *LMNA* was the strongest and only genotype independently associated with incident AF, conferring a 5-fold higher risk than in G(−) patients.

***TTN*: Structural Remodeling With Modest AF**

Although *TTN* variants are the most common cause of monogenic DCM, their associated AF burden is relatively modest. In our study, patients with *TTN* had a cumulative AF prevalence of 24%, similar to that in the G(−) group but approximately half that observed in *LMNA*. Across studies, AF prevalence in *TTN*-related cardiomyopathy ranges from 17% to 33%,^{9,22–24} consistently lower than that observed in *LMNA*-related cardiomyopathy. Importantly, although *TTN* was not significantly associated with incident AF, the wide CIs raise the possibility that the study was underpowered to detect modest *TTN*-specific associations.

Notably, in our study, patients with *TTN* had larger LA diameters than those with *LMNA*, yet their AF burden was substantially lower. This discordance suggests that LA size alone does not fully account for AF risk and reflects differing atrial pathophysiology across genotypes.

TTN-related AF may reflect coexisting intrinsic atrial arrhythmogenicity and secondary remodeling mechanisms. Reduced LA atrial strain independent of LV systolic dysfunction, together with cellular evidence of electrical instability and profibrotic signaling, supports an intrinsic atrial myopathy in *TTN*.^{23,25,26} Concurrently, LV systolic dysfunction, elevated filling pressures, and secondary

mitral regurgitation also promote secondary remodeling.^{25,27} The latter is consistent with the lower LVEF observed in *TTN* compared with *LMNA* in our cohort. A prospective study of 716 patients with DCM found similar LVEF, LA volumes, and AF prevalence in *TTN*-positive and *TTN*-negative patients,⁹ suggesting that the incremental effect of *TTN* may be modest once overt DCM is established. Importantly, this differs from early-onset AF, typically defined as onset <60 years without overt risk factors, in which *TTN* variants are enriched.^{28–30}

DSP: Low AF Risk Despite High Ventricular Arrhythmogenicity

DSP cardiomyopathy is increasingly recognized as an inflammatory cardiomyopathy with predominant LV involvement and frequent VA.⁷ However, AF is relatively uncommon in DSP, even during hot phases, with reported prevalence ranging from only 3% to 12%.^{23,25,26} This aligns with our findings, in which AF was observed in only 7% of patients with DSP.

Mechanistically, the low AF burden may relate to the regional myocardial involvement in DSP. Cardiac magnetic resonance and histopathologic studies consistently demonstrate fibrosis and inflammatory changes predominantly affecting the LV.^{7,31–33} In comparison, evidence for atrial involvement remains scarce, and our findings support the absence of substantial LA involvement to drive AF. These findings suggest that despite marked ventricular electrical instability, DSP does not manifest a primary atrial phenotype.

Traditional Clinical Risk Factors Remain Relevant for Incident AF in Genotyped DCM

Traditional clinical risk factors remain important contributors to incident AF despite genotype-based risk stratification. In our study, older age, male sex, and prior HF hospitalization were independently associated with incident AF risk, even after accounting for disease gene.

In contrast, cardiac structural and anthropometric variables, including body weight and LA diameter, did not demonstrate independent associations with incident AF after multivariable adjustment, despite being significant in univariable analyses. Both variables were correlated with sex, with males having greater body weight and larger LA diameters, suggesting that their apparent association with AF risk might reflect sex-related differences in body size and cardiac remodeling rather than independent effects in this cohort. Although obesity,³⁴ weight gain,³⁵ and atrial enlargement³⁶ are established AF risk factors, their lack of independent association in our study may reflect limited cumulative exposure or an earlier disease stage at the time of evaluation. Similarly, other common risk factors, such as diabetes and hypertension, showed nonsignificant associations with incident AF. This likely

reflects the relatively young age of our DCM cohort, which is below the average age of diagnosis for diabetes³⁷ and hypertension³⁸ in the general population.

Taken together, these findings suggest that apart from *LMNA*, incident AF in genotyped DCM is more likely to be influenced by conventional clinical factors, underscoring their continued relevance in AF risk assessment.

Prognostic Implications of Incident AF Development in Genotyped DCM

In our cohort, AF development during follow-up was independently associated with all evaluated adverse clinical outcomes after adjusting for genetic and clinical covariates. The strongest associations were observed for HF events and all-cause mortality, contributing most to the overall composite risk. These findings align with prior studies in ungenotyped DCM populations, in which AF has been consistently associated with worse prognosis.

Prior studies have associated incident AF with an ≈2- to 4-fold higher risk of all-cause mortality or urgent advanced HF therapies.^{1,39} Our findings extend this evidence to genotyped DCM, demonstrating that incident AF development remains an independent and incremental marker of adverse outcomes even after accounting for genetic substrate and clinical covariates, supporting its potential utility as a clinically actionable marker across genotyped DCM.

Stratified analyses revealed marked genotype-specific variation in clinical outcomes after AF onset. Patients with *LMNA*+AF exhibited the steepest risk trajectory, with double the risk of adverse events compared with the *TTN* cohort with AF, despite having a higher baseline LVEF. A similar trajectory was observed in a cohort study of 122 patients with *LMNA*, in which those presenting with atrial arrhythmia went on to develop new LV dysfunction (38%), new VA (30%), or end-stage HF or death (43%) during a median follow-up of 7 years.¹¹ Together, these findings reinforce that AF is a harbinger of broader myocardial involvement and adverse outcomes in patients with *LMNA*, rather than an isolated atrial abnormality. In contrast, patients with AF and either *TTN* or *G*(−) status had similarly lower cumulative event rates. Other less prevalent genotypes also show gene-specific AF associations, including AF risk comparable to *LMNA* in *RBM20*,⁴⁰ increased AF susceptibility with *SCN5A*,^{41,42} and familial AF clustering in *BAG3*.⁴³ In our study, small numbers precluded evaluation of these gene-specific associations.

Clinical Implications

Genetic testing is foundational in DCM management and is currently recommended in clinical guidelines for diagnosis, risk stratification, and family screening.⁴⁴ Our findings have important clinical implications for genotype-guided

AF care, given the strikingly elevated AF burden, incident risk, and adverse clinical course observed in patients with *LMNA*, compared with other patients with genotypes and *G*(−). These results support using *LMNA* status to inform surveillance intensity and therapeutic strategies.

In *LMNA*, where AF often signals early disease progression, our findings highlight the potential value of rhythm monitoring for detecting subclinical AF, including in asymptomatic *LMNA* carriers. This is particularly relevant given the significantly elevated thromboembolic events in *LMNA*+AF and the inadequacy of traditional risk scores (eg, CHA₂DS₂-VA) for accurately predicting this risk.⁴⁵ Furthermore, the onset of AF in *LMNA* may serve as a clinical signal to consider earlier evaluation for advanced HF therapies, given the accelerated disease progression that typically follows in this population.^{11,46} Both antiarrhythmic drugs and catheter ablation have limited efficacy for AF in *LMNA*.^{20,47} Collectively, these observations provide rationale for prospective evaluation of intensified AF surveillance and genotype-guided management in *LMNA*.

In contrast, patients with non-*LMNA* variants have a relatively lower AF burden, with those carrying *TTN* showing comparatively better outcomes than those with *LMNA*. Recognizing this distinction is essential to avoid overestimation of AF risk in these subgroups and to tailor monitoring and management strategies. Furthermore, early AF rhythm control strategies, particularly catheter ablation, seem effective in the non-*LMNA* group.⁴⁷ In a study of patients with AF, recurrence rates after AF ablation among patients with *TTN* were comparable to those in *G*(−) controls.¹³

Limitations

Although the SHaRe-DCM consortium represents one of the largest genotyped DCM cohorts, center-level heterogeneity inherent to this multicenter registry may have influenced AF ascertainment and outcomes. All Cox models were therefore stratified by enrolling SHaRe site, but residual heterogeneity cannot be excluded.

Certain genotypes (eg, sarcomeric genes) were underpowered for individual analysis, necessitating grouping to maintain statistical validity. AF ascertainment was based on clinical documentation and may underestimate the true burden of paroxysmal or subclinical episodes, because rhythm monitoring and cardiac device use were not uniform across sites. In addition, follow-up duration varied across the AF groups, with shorter follow-up in the no AF group, potentially contributing to underascertainment in our cohort.

The lack of higher AF rates in *G*(+) patients with non-*LMNA* variants compared with *G*(−) patients may reflect referral and enrollment bias in SHaRe-DCM, which preferentially includes patients with phenotypes suggestive of genetic or arrhythmogenic cardiomyopathies.

Consequently, the comparison between *TTN* and *G*(−) patients may underestimate the true *TTN*-specific AF risk relative to unselected DCM populations. This limits generalizability to broader DCM populations but preserves the validity of genotype-stratified comparisons within the registry. AF preceding DCM in the *LMNA* cohort may partly reflect lead-time bias, because DCM onset was defined by the first documented LVEF <50%. Clinically relevant covariates, including guideline-directed medical therapy, anticoagulation, and structural abnormalities such as mitral valve disease, were not systematically collected, precluding adjustment for related confounding.

The retrospective, multicenter design led to variable data completeness across sites, addressed through imputation and targeted manual review of medical records where appropriate. Given the number of genotype subgroups and outcomes evaluated, and the absence of formal correction for multiple comparisons, these findings should be interpreted as exploratory and hypothesis-generating. Although these limitations introduce the potential for residual confounding and limit causal inference, the consistency of our findings with prior studies supports the clinical relevance.

Conclusions

AF in DCM is heterogeneous with genotype-specific variation. *LMNA* variants confer the highest burden and risk of incident AF and are associated with significantly worse outcomes compared with other genotypes and *G*(−) patients. Recognizing and integrating these genotype-specific differences can allow for more personalized and effective care strategies in DCM. Future studies are warranted to develop and validate genotype-specific AF risk models to further refine clinical decision-making and improve outcomes in DCM.

ARTICLE INFORMATION

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Supplemental Material

Figure S1
Tables S1–S6

REFERENCES

- Aleksova A, Merlo M, Zecchin M, Sabbadini G, Barbati G, Vitrella G, Di Lenarda A, Sinagra G. Impact of atrial fibrillation on outcome of patients with idiopathic dilated cardiomyopathy: data from the Heart Muscle Disease Registry of Trieste. *Clin Med Res*. 2010;8:142–149. doi: 10.3121/cmr.2010.908
- Buckley BJR, Harrison SL, Gupta D, Fazio-Eynullayeva E, Underhill P, Lip GYH. Atrial fibrillation in patients with cardiomyopathy: prevalence and clinical outcomes from real-world data. *J Am Heart Assoc*. 2021;10:e021970. doi: 10.1161/jaha.121.021970
- Ródenas-Alesina E, Lozano-Torres J, Tobías-Castillo PE, Badia-Molins C, Calvo-Barceló M, Vila-Olives R, Casas-Masnou G, San Emeterio AO, Soriano-Colomé T, Fernández-Galera R, et al. Risk of stroke and incident atrial fibrillation in patients in sinus rhythm with nonischemic dilated cardiomyopathy. *Am J Cardiol*. 2024;233:11–18. doi: 10.1016/j.amjcard.2024.09.024
- Antonopoulos AS, Kasiakogias A, Kouroutzoglou A, Touloupaki M, Briassoulis A, Papatheodorou E, Rigopoulos AG, Antonakaki D, Laina A, Tsioufis K, et al. Atrial fibrillation burden and management in cardiomyopathies: current evidence and unmet needs. *Trends Cardiovasc Med*. 2025;35:284–293. doi: 10.1016/j.tcm.2025.01.007
- Santhanakrishnan R, Wang N, Larson MG, Magnani JW, McManus DD, Lubitz SA, Ellinor PT, Cheng S, Vasan RS, Lee DS, et al. Atrial fibrillation

- begets heart failure and vice versa: temporal associations and differences in preserved versus reduced ejection fraction. *Circulation*. 2016;133:484–492. doi: 10.1161/CIRCULATIONAHA.115.018614
6. Elliott AD, Middeldorp ME, Van Gelder IC, Albert CM, Sanders P. Epidemiology and modifiable risk factors for atrial fibrillation. *Nat Rev Cardiol*. 2023;20:404–417. doi: 10.1038/s41569-022-00820-8
 7. Smith ED, Lakdawala NK, Papoutsidakis N, Aubert G, Mazzanti A, McCanta AC, Agarwal PP, Arscott P, Dellefave-Castillo LM, Vorovich EE, et al. Desmoplakin cardiomyopathy, a fibrotic and inflammatory form of cardiomyopathy distinct from typical dilated or arrhythmogenic right ventricular cardiomyopathy. *Circulation*. 2020;141:1872–1884. doi: 10.1161/CIRCULATIONAHA.119.044934
 8. Nuzzi V, Cannatà A, Manca P, Castrichini M, Barbati G, Aleksova A, Fabris E, Zecchin M, Merlo M, Boriani G, et al. Atrial fibrillation in dilated cardiomyopathy: outcome prediction from an observational registry. *Int J Cardiol*. 2021;323:140–147. doi: 10.1016/j.ijcard.2020.08.062
 9. Tayal U, Newsome S, Buchan R, Whiffin N, Halliday B, Lota A, Roberts A, Baksi AJ, Voges I, Midwinter W, et al. Phenotype and clinical outcomes of titin cardiomyopathy. *J Am Coll Cardiol*. 2017;70:2264–2274. doi: 10.1016/j.jacc.2017.08.063
 10. Hey TM, Rasmussen TB, Madsen T, Aagaard MM, Harbo M, Mølgaard H, Møller JE, Eiskjær H, Mogensen J. Pathogenic RBM20-variants are associated with a severe disease expression in male patients with dilated cardiomyopathy. *Circ Heart Fail*. 2019;12:e005700. doi: 10.1161/CIRCHEARTFAILURE.118.005700
 11. Kumar S, Baldinger SH, Gandjbakhch E, Maury P, Sellal JM, Androulakis AFA, Waintraub X, Charron P, Rollin A, Richard P, et al. Long-term arrhythmic and nonarrhythmic outcomes of lamin A/C mutation carriers. *J Am Coll Cardiol*. 2016;68:2299–2307. doi: 10.1016/j.jacc.2016.08.058
 12. Ortiz-Genga MF, Cuenca S, Dal Ferro M, Zorio E, Salgado-Aranda R, Climent V, Padrón-Barthe L, Duro-Aguado I, Jiménez-Jáimez J, Hidalgo-Olivares VM, et al. Truncating FLNC mutations are associated with high-risk dilated and arrhythmogenic cardiomyopathies. *J Am Coll Cardiol*. 2016;68:2440–2451. doi: 10.1016/j.jacc.2016.09.927
 13. Virk ZM, El-Harasis MA, Yoneda ZT, Anderson KC, Sun L, Quintana JA, Murphy BS, Laws JL, Davogusto GE, O'Neill MJ, et al. Clinical characteristics and outcomes with atrial fibrillation and pathogenic TTN variants. *JACC Clin Electrophysiol*. 2024;10:2445–2457. doi: 10.1016/j.jacep.2024.07.029
 14. Marcus FI, McKenna WJ, Sherrill D, Basso C, Bauce B, Bluemke DA, Calkins H, Corrado D, Cox MGRJ, Daubert JP, et al. Diagnosis of arrhythmogenic right ventricular cardiomyopathy/dysplasia: proposed modification of the Task Force Criteria. *Eur Heart J*. 2010;31:806–814. doi: 10.1093/eurheartj/ehq025
 15. Miller DT, Lee K, Abul-Husn NS, Amendola LM, Brothers K, Chung WK, Gollob MH, Gordon AS, Harrison SM, Hershberger RE, et al; ACMG Secondary Findings Working Group. Electronic address: documents@acmg.net. ACMG SF v3.2 list for reporting of secondary findings in clinical exome and genome sequencing: a policy statement of the American College of Medical Genetics and Genomics (ACMG). *Genet Med*. 2023;25:100866. doi: 10.1016/j.gim.2023.100866
 16. ClinGen curated genes. Accessed May 8, 2025. <https://search.clinicalgenome.org/kb/genes/curations?page=1&size=25&search=>
 17. Balakrishnan ID, Lakdawala NK. Contemporary insights into LMNA cardiomyopathy. *Curr Cardiol Rep*. 2025;27:40. doi: 10.1007/s11886-025-02195-x
 18. Topriceanu CC, Alfari H, Hughes AD, Shiwan H, Chan F, Mohiddin SA, Moyle W, Steeds RP, O'Brien B, Vowinkel J, et al. The atrial and ventricular myocardial proteome of end-stage lamin heart disease. *Acta Myol*. 2023;42:43–52. doi: 10.36185/2532-1900-339
 19. Zhang Y, Lin Y, Zhang Y, Wang Y, Li Z, Zhu Y, Liu H, Ju W, Cui C, Chen M. Familial atrial myopathy in a large multigenerational heart-hand syndrome pedigree carrying an LMNA missense variant in rod 2B domain (p.R335W). *Heart Rhythm*. 2022;19:466–475. doi: 10.1016/j.hrthm.2021.11.022
 20. Chauvel R, Derval N, Duchateau J, Denis A, Tixier R, Welte N, André C, Ramirez FD, Nakashima T, Nakatani Y, et al. Persistent atrial fibrillation ablation in cardiac laminopathy: electrophysiological findings and clinical outcomes. *Heart Rhythm*. 2021;18:1115–1121. doi: 10.1016/j.hrthm.2021.03.040
 21. Tremblay-Gravel M, Ichimura K, Picard K, Kawano Y, Dries AM, Haddad F, Lakdawala NK, Wheeler MT, Parikh M. Intrinsic atrial myopathy precedes left ventricular dysfunction and predicts atrial fibrillation in lamin A/C cardiomyopathy. *Circ Genomic Precis Med*. 2023;16:20–28. doi: 10.1161/CIRCGEN.121.003480
 22. Akhtar MM, Lorenzini M, Cicerchia M, Ochoa JP, Hey TM, Sabater Molina M, Restrepo-Cordoba MA, Dal Ferro M, Stolfo D, Johnson R, et al. Clinical phenotypes and prognosis of dilated cardiomyopathy caused by truncating variants in the TTN gene. *Circ Heart Fail*. 2020;13:e006832–e006508. doi: 10.1161/CIRCHEARTFAILURE.119.006832
 23. Henkens MTHM, Raafs AG, Vanloon T, Vos JL, Vandenwijngaard A, Brunner HG, Krapels IPC, Knackstedt C, Gerretsen S, Hazebroek MR, et al. Left atrial function in patients with titin cardiomyopathy. *J Card Fail*. 2024;30:51–60. doi: 10.1016/j.cardfail.2023.05.013
 24. Schiabor Barrett KM, Cirulli ET, Bolze A, Rowan C, Elhanan G, Grzymalski JJ, Lee W, Washington NL. Cardiomyopathy prevalence exceeds 30% in individuals with TTN variants and early atrial fibrillation. *Genet Med*. 2023;25:100012. doi: 10.1016/j.gim.2023.100012
 25. De Castro D, Fatkin D, Rodriguez-Rubio E, Haddad F, Hernández-Terciado F, Purohit A, Bagherzadeh SP, Nair P, Rad A, Krysov VA, et al. Truncating variants in TTN are associated with primary atrial myopathy. *J Am Heart Assoc*. 2026;15:eJAH2025047740T. doi: 10.1161/JAHA.125.047740
 26. Cumberland MJ, Dasí A, Sontayananon N, Marcus AD, Qin AR, Riebel L, Euchner J, Azad AJ, Hall C, O'Shea C, et al. A titin truncating variant linked to atrial fibrillation increases atrial profibrotic signalling and cholinergic sensitivity. *Cardiovasc Res*. 2026;122:1206–1223. doi: 10.1093/cvr/cvag112
 27. Inciardi RM, Bonelli A, Biering-Sorensen T, Cameli M, Pagnesi M, Lombardi CM, Solomon SD, Metra M. Left atrial disease and left atrial reverse remodelling across different stages of heart failure development and progression: a new target for prevention and treatment. *Eur J Heart Fail*. 2022;24:959–975. doi: 10.1002/ehfj.2562
 28. Rudaka I, Vilne B, Isakova J, Kalejs O, Gailite L, Rots D. Genetic basis of early onset atrial fibrillation in patients without risk factors. *J Cardiovasc Dev Dis*. 2023;10:104. doi: 10.3390/jcdd10030104
 29. Lazarte J, Laksman ZW, Wang J, Robinson JF, Dron JS, Leach E, Liew J, McIntyre AD, Skanes AC, Gula LJ, et al. Enrichment of loss-of-function and copy number variants in ventricular cardiomyopathy genes in "lone" atrial fibrillation. *Europace*. 2021;23:844–850. doi: 10.1093/eurpace/euraa421
 30. Yoneda ZT, Anderson KC, Quintana JA, O'Neill MJ, Sims RA, Glazer AM, Shaffer CM, Crawford DM, Stricker T, Ye F, et al. Early-onset atrial fibrillation and the prevalence of rare variants in cardiomyopathy and arrhythmia genes. *JAMA Cardiol*. 2021;6:1371–1379. doi: 10.1001/jamacardio.2021.3370
 31. Bariani R, Cason M, Rigato I, Cipriani A, Celeghin R, De Gaspari M, Bueno Marinas M, Mattesi G, Pergola V, Rizzo S, et al. Clinical profile and long-term follow-up of a cohort of patients with desmoplakin cardiomyopathy. *Heart Rhythm*. 2022;19:1315–1324. doi: 10.1016/j.hrthm.2022.04.015
 32. Di Lorenzo F, Marchionni E, Ferradini V, Latini A, Pezzoli L, Martino A, Romeo F, Iorio A, Bianchi S, Iacone M, et al. DSP-related cardiomyopathy as a distinct clinical entity? Emerging evidence from an Italian cohort. *Int J Mol Sci*. 2023;24:2490. doi: 10.3390/ijms24032490
 33. De Frutos F, Ochoa JP, Fernández AI, Gallego-Delgado M, Navarro-Peñalver M, Casas G, Basurte MT, Larranaga-Moreira JM, Mogollón MV, Robles-Mezcua A, et al. Late gadolinium enhancement distribution patterns in non-ischaemic dilated cardiomyopathy: genotype-phenotype correlation. *Eur Heart J Cardiovasc Imaging*. 2023;25:75–85. doi: 10.1093/ehjci/jead184
 34. Sha R, Baines O, Hayes A, Tompkins K, Kalla M, Holmes AP, O'Shea C, Pavlovic D. Impact of obesity on atrial fibrillation pathogenesis and treatment options. *J Am Heart Assoc*. 2024;13:e032277. doi: 10.1161/JAHA.123.032277
 35. Du Y, Qi L, Borné Y, Sonestedt E. Adulthood weight changes, body mass index in youth, genetic susceptibility and risk of atrial fibrillation: a population-based cohort study. *BMC Med*. 2024;22:345. doi: 10.1186/s12916-024-03565-y
 36. Olsen JF, Møgelvang R, Jensen GB, Jensen JS, Biering-Sørensen T. Relationship between left atrial functional measures and incident atrial fibrillation in the general population: the Copenhagen City Heart Study. *JACC Cardiovasc Imaging*. 2019;12:981–989. doi: 10.1016/j.jcmg.2017.12.016
 37. Huang X, Lee K, Wang MC, Shah NS, Khan SS. Age at diagnosis of hypertension by race and ethnicity in the US from 2011 to 2020. *JAMA Cardiol*. 2022;7:986–987. doi: 10.1001/jamacardio.2022.2345
 38. Wang MC, Shah NS, Carnethon MR, O'Brien MJ, Khan SS. Age at diagnosis of diabetes by race and ethnicity in the United States from 2011 to 2018. *JAMA Intern Med*. 2021;181:1537–1539. doi: 10.1001/jamainternmed.2021.4945
 39. Dziewięcka E, Gliniak M, Winiarczyk M, Karapetyan A, Wiśniowska-Śmiełek S, Karabinowska A, Holcman K, Kostkiewicz M, Hlawaty M, Leśniak-Sobielga A, et al. The burden of atrial fibrillation and its prognostic value in patients with dilated cardiomyopathy. *Kardiologia Pol*. 2020;78:37–44. doi: 10.33963/KP.15046
 40. Tukker M, Te Rijdt W, Amin AS, Morris-Rosendahl DJ, Hirsch A, Ben-Haim Y, Houweling AC, Varnava A, Behr ER, Hajdarpasic A, et al. High incidence of malignant arrhythmias and heart failure in patients with

- RBM20-associated cardiomyopathy: a multicenter cohort study and review of the literature. *Eur Heart J*. 2024;45(suppl 1):ehae666.651. doi: 10.1093/eurheartj/ehae666.651
41. Darbar D, Kannankeril PJ, Donahue BS, Kucera G, Stubblefield T, Haines JL, George AL, Roden DM. Cardiac sodium channel (SCN5A) variants associated with atrial fibrillation. *Circulation*. 2008;117:1927–1935. doi: 10.1161/CIRCULATIONAHA.107.757955
 42. Pierre M, Djemai M, Pouliot V, Poulin H, Gollob MH, Chahine M. Exploring SCN5A variants associated with atrial fibrillation in atrial cardiomyocytes derived from human induced pluripotent stem cells: a characterization study. *Heart Rhythm*. 2025;22:1574–1587. doi: 10.1016/j.hrthm.2024.09.013
 43. Mėlinytė-Ankudavičė K, Šukys M, Plisienė J, Jurkevičius R, Ereminienė E. Genotype-phenotype correlation in familial BAG3 mutation dilated cardiomyopathy. *Genes*. 2022;13:363. doi: 10.3390/genes13020363
 44. Arbelo E, Protonotarios A, Gimeno JR, Arbustini E, Barriales-Villa R, Basso C, Bezzina CR, Biagini E, Blom NA, de Boer RA, et al; ESC Scientific Document Group. 2023 ESC guidelines for the management of cardiomyopathies [published correction appears in *Eur Heart J*: published online August 17, 2026. doi: 10.1093/eurheartj/ehag645]. *Eur Heart J*. 2023;44:3503–3626. doi: 10.1093/eurheartj/ehad194. <https://academic.oup.com/eurheartj/article/44/37/3503/7246608>
 45. Balakrishnan I, Salbinski L, Nozica N, Andrews E, Sylvester KW, Claggett BL, Connors JM, Tedrow U, Lakdawala NK. Atrial fibrillation and systemic embolic events in patients with LMNA cardiomyopathy. *J Am Coll Cardiol*. 2025;86:404–407. doi: 10.1016/j.jacc.2025.05.039
 46. Hasselberg NE, Haland TF, Saberniak J, Brekke PH, Berge KE, Leren TP, Edvardsen T, Haugaa KH. Lamin A/C cardiomyopathy: young onset, high penetrance, and frequent need for heart transplantation. *Eur Heart J*. 2018;39:853–860. doi: 10.1093/eurheartj/ehx596
 47. El-Harasis MA, Yoneda ZT, Anderson KC, Ye F, Quintana JA, Martinez-Parachini JR, Jackson GG, Varghese BT, Crawford DM, Sun L, et al. Ablation for atrial fibrillation in patients with rare pathogenic variants in cardiomyopathy and arrhythmia genes. *JACC Clin Electrophysiol*. 2024;10:2359–2370. doi: 10.1016/j.jacep.2024.06.035



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